2013 Homework 1

MOS theory, multiplexers and static CMOS

The original printed prompt is preserved at `../assignments/homework1.pdf`. In Question 4(a), there are two nested overbars. All diagrams use PMOS gate bubbles; NMOS bodies are grounded and PMOS bodies connect to V_{DD}.

1. Devices that cannot conduct in saturation

The ground-side NMOS is M_1, the output-side NMOS is M_2; the supply-side PMOS is M_4, and the output-side PMOS is M_3. Let u be the NMOS intermediate voltage. Conducting saturation of M_1 would require $u \geq V_{in} - V_{tn}$. Then

$$V_{GS,2} = V_{in} - u \leq V_{tn},$$

so M_2 could not carry positive channel current. Similarly, if v is the PMOS intermediate voltage, saturation of M_4 implies

$$V_{DD} - v \geq V_{DD} - V_{in} - |V_{tp}| \implies V_{SG,3} = v - V_{in} \leq |V_{tp}|.$$

Thus M_3 would be off. The requested devices are

$$\boxed{M_1 \text{ and } M_4, \text{ for nonzero conducting stack current.}}$$

The zero-current cutoff boundary is excluded from the conducting-region claim.

2. Two series MOSFETs have length $L_1 + L_2$

Use the ideal long-channel square-law model, common gate and width W, equal thresholds, no body effect and no channel-length modulation. Let $U = V_{in} - V_t > 0$, $v = V_{DS}$ for the whole stack, and $\kappa = \mu C_{ox} W$. Define $\phi_U(q) = Uq - q^2/2$. When both devices are linear, their equal currents satisfy

$$\frac{IL_1}{\kappa} = \phi_U(u), \qquad \frac{IL_2}{\kappa} = \phi_U(v) - \phi_U(u).$$

Adding eliminates the unknown intermediate node:

$$\boxed{I = \frac{\kappa}{L_1 + L_2}(Uv - v^2/2), \quad v < U.}$$

This is exactly the linear characteristic of one device with length $L_1 + L_2$.

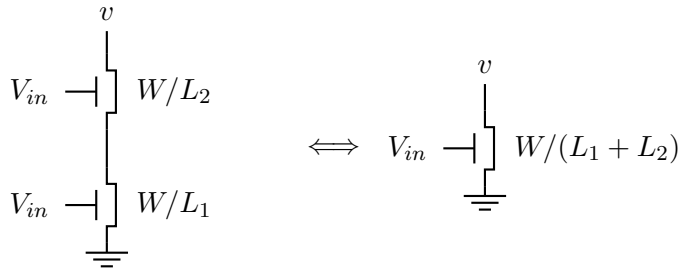
For $v \geq U$, the upper device saturates and the lower remains linear:

$$\frac{\kappa}{L_1}(Uu - u^2/2) = \frac{\kappa}{2L_2}(U-u)^2.$$

Set $q = U - u$. Then $L_2(U^2 - q^2) = L_1 q^2$ and therefore

$$q = U\sqrt{\frac{L_2}{L_1+L_2}}, \qquad \boxed{I = \frac{\kappa U^2}{2(L_1+L_2)}, \quad v \geq U.}$$

The equivalent device also matches the saturation current. For $L_1 = L_2 = L$, the effective length is $2L$ and the aspect ratio is $W/(2L)$. This is a static ideal-model equivalence; diffusion capacitance, body effect and short-channel physics prevent a claim of identical physical transient behavior.



Reference comparison. The supplied `MOS prove.pdf`, problems 6.14 and 6.15 on pages 5–6, supports the cutoff contradiction and linear length-addition calculation. The derivation above additionally proves the saturation case and allows unequal lengths. The companion program checks 80,800 operating points across 16 unequal-length pairs; the maximum normalized current error is 3.89×10^{-16}.

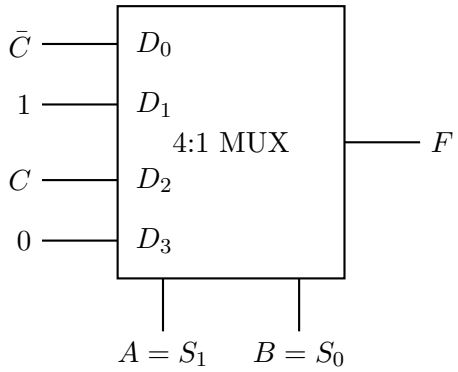
3. Implement the function with one 4:1 multiplexer

Read the individual bars in the original expression:

$$\boxed{F = \bar{A}(B + \bar{C}) + A\bar{B}C.}$$

Choose select lines $(S_1, S_0) = (A, B)$. The four cofactors are

A	B	MUX input	Value
0	0	D_0	$\bar{C}$
0	1	D_1	1
1	0	D_2	C
1	1	D_3	0



Generate $\bar{C}$ with one CMOS inverter if it is not already available. This uses one 4:1 multiplexer, as requested, rather than an unrestricted 2:1 tree.

A	B	C	F	A	B	C	F
0	0	0	1	1	0	0	0
0	0	1	0	1	0	1	1
0	1	0	1	1	1	0	0
0	1	1	1	1	1	1	0

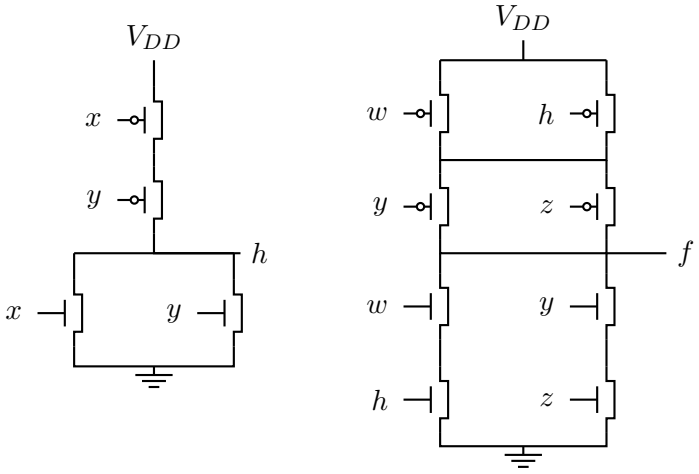
All eight cases are checked independently against the printed Boolean expression. No complete local submitted HW1 answer accompanies the prompt; the MOS proof attachment covers Questions 1 and 2 only. The remaining designs are derived here.

4(a). Static CMOS with two nested overbars

The outer bar covers the whole sum, and the inner bar covers $(x + y)$:

$$\boxed{f = \overline{w\,\overline{(x+y)} + yz.}}$$

First generate $h = \overline{x+y}$ with a four-transistor NOR. Then use an eight-transistor AOI22 gate: its PDN is (n_w series n_h) in parallel with (n_y series n_z); the dual PUN is ($p_w \parallel p_h$) in series with ($p_y \parallel p_z$). Only true *external* inputs are required.

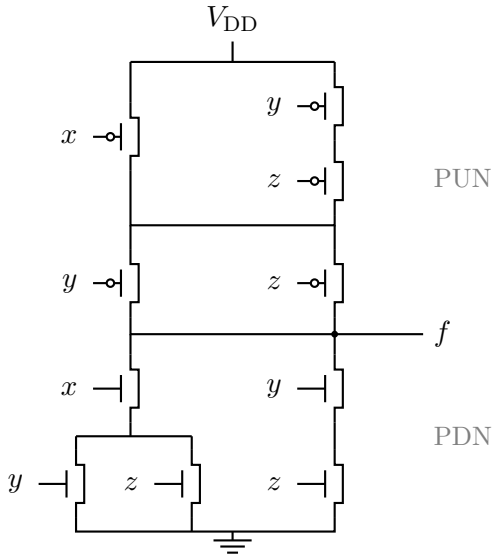


The two diagrams share the labeled h signal. Total: $4 + 8 = \boxed{12}$ MOSFETs. The checker solves the NOR's actual switch network before evaluating the AOI22 network, then tests all 16 input words against the nested expression. Removing the inner bar would produce a different function; it is not a permissible reuse of a later year's AOI diagram.

4(b). Complement of three-input majority

$$f = \bar{x}\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z} = \overline{xy + xz + yz} = \overline{x(y+z) + yz}.$$

The PDN is n_x in series with $(n_y \parallel n_z)$, in parallel with n_y in series with n_z. The PUN is its series/parallel dual.



Minority gate: 5 NMOS + 5 PMOS.

It is high for input Hamming weight 0 or 1 and low for weight 2 or 3. No external complemented inputs are needed: PMOS gate bubbles supply the complementary switch-enable condition. All eight states have exactly one conducting rail network. **Verification.** Run `python verify.py`. All 32 Boolean/switch cases and all 80,800 long-channel operating points pass. Results are preserved in `truth_tables.csv` and `verification.json`. These are ideal logic and model-equation checks, not physical SPICE timing measurements.